

Experiment 3: Comparison of Butterworth and Chebyshev Type-I Filters

MATLAB Code:

```
clc;
clear;
close all;

%% Specifications
Fs = 2000;
Rp = 1;
Rs = 40;

Wp = [300 600]/(Fs/2);
Ws = [200 700]/(Fs/2);

%% Save Command Window Output
diary('Exp3_1_Output.txt');

%% Butterworth Filter
[n1,W1] = buttord(Wp,Ws,Rp,Rs);
[b1,a1] = butter(n1,W1,'bandpass');

%% Chebyshev Type-I Filter
[n2,W2] = cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2] = cheby1(n2,Rp,W2,'bandpass');

%% Butterworth Frequency Response
figure(1);
freqz(b1,a1);
title('Butterworth Filter');

%% Chebyshev Frequency Response
figure(2);
freqz(b2,a2);
title('Chebyshev Type-I Filter');

%% Group Delay
figure(3);
grpdelay(b1,a1);
hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');
title('Group Delay');

%% Pole-Zero Plot
figure(4);

subplot(1,2,1);
zplane(b1,a1);
title('Butterworth');

subplot(1,2,2);
zplane(b2,a2);
title('Chebyshev-I');

%% Impulse Response
figure(5);

subplot(2,1,1);
impz(b1,a1);
title('Butterworth Impulse');

subplot(2,1,2);
impz(b2,a2);
title('Chebyshev Impulse');

%% Pole Magnitude
figure(6);
```

```

subplot(1,2,1);
stem(abs(roots(a1)), 'filled');
title('Butterworth Pole Magnitude');

subplot(1,2,2);
stem(abs(roots(a2)), 'filled');
title('Chebyshev Pole Magnitude');

%% Display Filter Orders
disp(['Butterworth Order = ', num2str(n1)]);
disp(['Chebyshev Order = ', num2str(n2)]);

%% Stop Saving Command Window
diary off;

%% Save Figures (High Quality - 300 DPI)

exportgraphics(figure(1), 'Exp3_1_Butterworth_Filter.png', 'Resolution', 300);
exportgraphics(figure(2), 'Exp3_1_Chebyshev_Filter.png', 'Resolution', 300);
exportgraphics(figure(3), 'Exp3_1_Group_Delay.png', 'Resolution', 300);
exportgraphics(figure(4), 'Exp3_1_Pole_Zero_Plot.png', 'Resolution', 300);
exportgraphics(figure(5), 'Exp3_1_Impulse_Response.png', 'Resolution', 300);
exportgraphics(figure(6), 'Exp3_1_Pole_Magnitude.png', 'Resolution', 300);

%% Completion Message
disp(' ');
disp('=====');
disp('Experiment 3.1 completed successfully. ');
disp('Files Saved: ');
disp('Exp3_1_Output.txt');
disp('Exp3_1_Butterworth_Filter.png');
disp('Exp3_1_Chebyshev_Filter.png');
disp('Exp3_1_Group_Delay.png');
disp('Exp3_1_Pole_Zero_Plot.png');
disp('Exp3_1_Impulse_Response.png');
disp('Exp3_1_Pole_Magnitude.png');
disp('=====');

```

Output:

Butterworth Order = 9
Chebyshev Order = 5

Output Figures:





